



11527 - Granulation in Color: Tracing Stellar Flicker Across Wavelengths

Cycle: 5, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Jeff A. Valenti (PI)	Space Telescope Science Institute
Dr. Ana Glidden (CoI) (CoPI)	Massachusetts Institute of Technology
Dr. Alexander I. Shapiro (CoI) (ESA Member)	Max Planck Institute for Solar System Research
Dr. Nadiia Kostogryz (CoI) (ESA Member)	Max Planck Institute for Solar System Research
Dr. Arika Egan (CoI)	The Johns Hopkins University Applied Physics Laboratory
Dr. Nestor Espinoza (CoI)	Space Telescope Science Institute
Dr. Mans Holmberg (CoI)	Space Telescope Science Institute
Prof. Sara Seager (CoI)	Massachusetts Institute of Technology
Dr. Greg Kopp (CoI)	University of Colorado at Boulder

OBSERVATIONS

ABSTRACT

High-precision JWST light curves have provided unprecedented insights into stellar physics. Daily variations in these light curves probe rotational modulation of surface magnetic features, while minute-scale oscillation signals probe global properties and internal structure. In contrast, the diagnostic potential of intermediate-timescale variability, tens of minutes to a few hours, often termed stellar flicker—has remained unclear. This is primarily due to the absence of accurate models for flicker and its dependence on fundamental stellar parameters. The situation has recently changed due to the tremendous progress in our understanding of stellar near-surface convection. Recent simulations with the 3D radiative magnetohydrodynamics code MURaM reproduced the granulation-driven variability of the Total Solar Irradiance (TSI) and predicted a strong dependence of the stellar brightness variability on wavelength, fundamental stellar parameters, and surface magnetic field strength. We propose to utilize the unprecedented spectral sensitivity and photometric precision of JWST to measure wavelength-dependent stellar flicker for three solar-like stars spanning spectral types F6V, G2V, and G7V, comparing observations to MURaM simulations. In particular, we will test whether the influence

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of surface magnetic fields on stellar flicker, predicted by MURaM simulations, can be leveraged to measure small-scale magnetic fields that elude traditional diagnostics yet still contaminate exoplanet atmospheric measurements. This effort will test and refine our understanding of stellar near-surface convection, potentially opening a new window for stellar magnetic field diagnostics.

OBSERVING DESCRIPTION